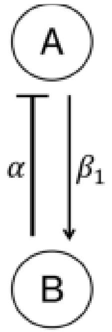
A Integral feedback model

$\frac{dA}{dt} = \alpha\,(\mu - B)$

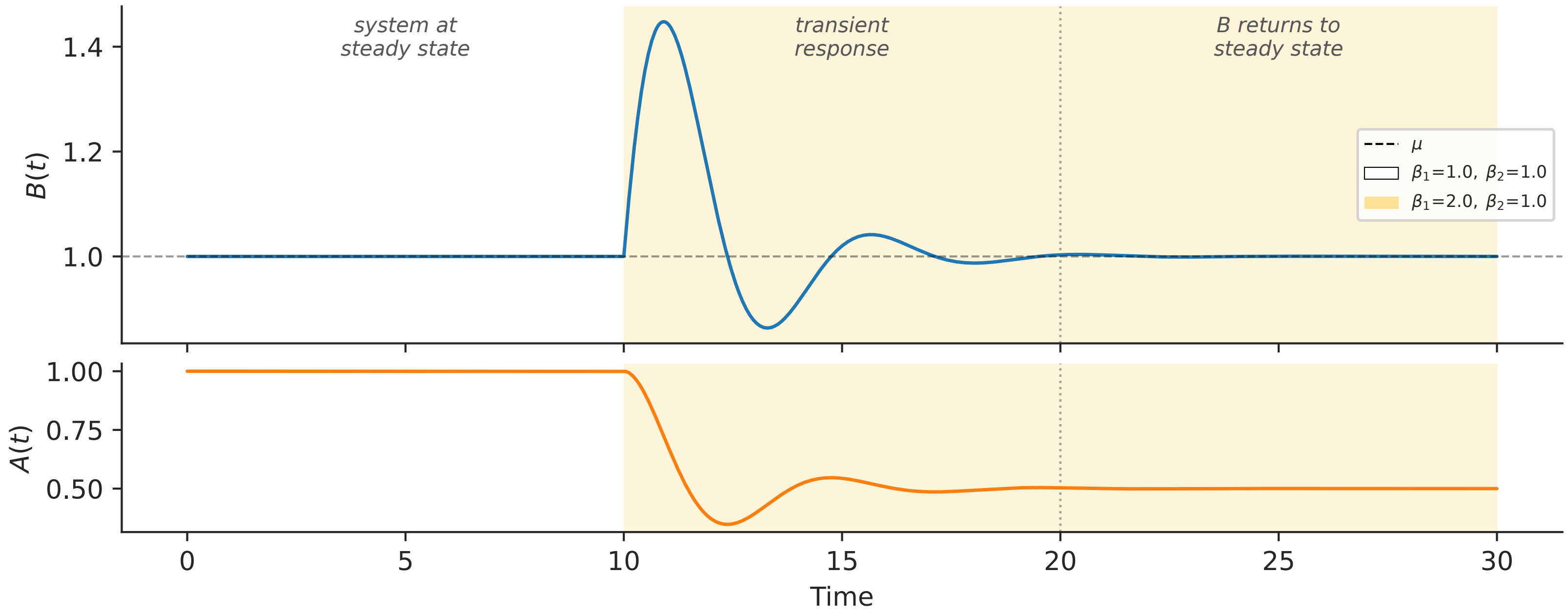
$\frac{dB}{dt} = \beta_1 A - \beta_2 B$

$B_{ss} = \mu$



Steady-state of B depends only on μ, and is thus robust to β_1 and β_2.

B Integral feedback ensures B converges to μ after parameter change

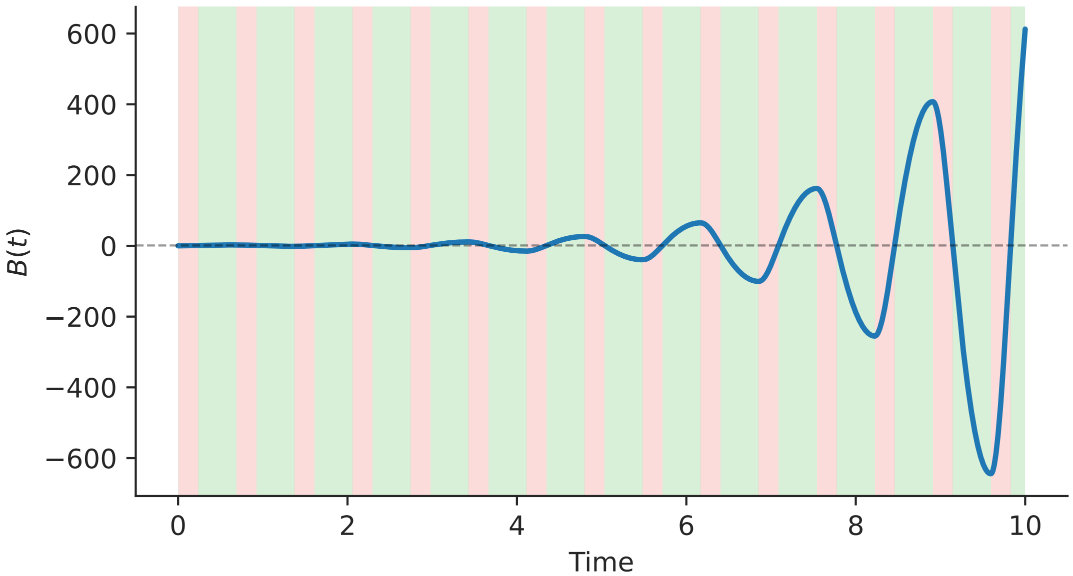


C Integral feedback model is equivalent to a mass on a spring

$\frac{d^2B}{dt^2} + \beta_2\frac{dB}{dt} + \alpha\beta_1 B = \alpha\beta_1\mu$

mass $m = 1$
damping $c = \beta_2$
spring constant $k = \alpha\beta_1$

D Adaptive strategy drives growth in AB model



E Calcium model with integral feedback is also equivalent to a mass on a spring

$P = k_0(R_0 - Ca)$

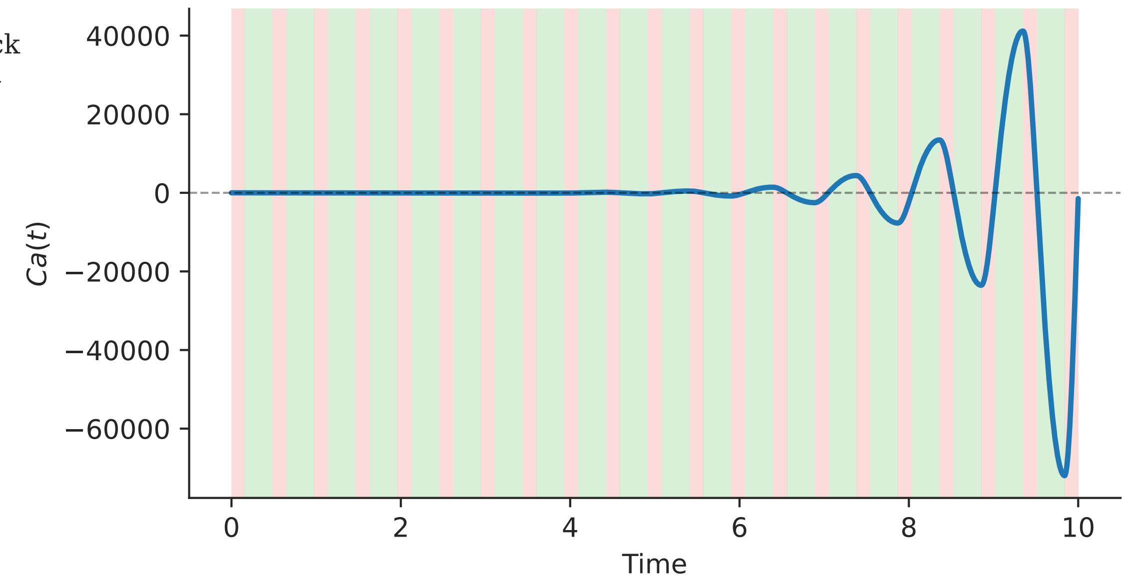
$\dot{Ca} = k_1 P + k_2 D$

$\dot{D} = k_3 P$

$Ca_{ss} = R_0$

spring constant $k = k_2 k_3 k_0$

F Adaptive strategy drives growth in calcium model



G Adaptive strategy that amplifies the motion of a mass on a spring